

Transport versus phase in the local gauge expansion at a simple transport ramification point

OpenAI working note

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Abstract. We determine the next symbolic term beyond the geometry-only coefficient K_1^{geom} in the local gauge expansion near a simple transport ramification point for Type-1, $N = 3$ Landau–Zener systems. In the natural *linear Langer/WKB normalization*, the determinant-one local gauge factors into a transport/half-form part and a phase remainder,

$$\widehat{G}_v(t; H) = \widehat{G}_v^{\text{tr}}(t) \exp(\Phi_v^{\text{ph}}(t; H) \sigma_3), \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

with

$$\widehat{G}_v^{\text{tr}}(t) = \widehat{G}_v^{\text{lead}} \exp(K_1^{\text{geom}}(v)t \sigma_3 + K_3^{\text{geom}}(v)t^3 \sigma_3 + O(t^5)),$$

and

$$\Phi_v^{\text{ph}}(t; H) = \Xi_5(H)t^5 + O(t^7).$$

Thus the first two nontrivial transport coefficients are geometry-only, while the first genuine member dependence appears at order t^5 . Closed symbolic formulas are given for $K_1^{\text{geom}}(v)$, $K_3^{\text{geom}}(v)$, and $\Xi_5(H)$, and an explicit jump-law corollary is derived for the active upper- and lower-half-plane selector blocks.

1 Setup and notation

Fix a real Type-1, $N = 3$ commuting family. The shared rational map and transport normalization are

$$u(\lambda) = \frac{n(\lambda)}{p(\lambda)}, \quad \Sigma^{(2)}(\lambda) = \frac{W_4(\lambda)}{p(\lambda)^2}.$$

Let $v \in \mathbb{P}_\lambda^1$ be a *simple transport ramification point*, so that

$$W_4(v) = 0, \quad W_4'(v) \neq 0, \quad u_* = u(v).$$

On the local double cover we write

$$u = u_* + t^2.$$

The chosen commuting-family member H enters through the phase carrier

$$\Delta_H(\lambda) = \mu(\lambda) \frac{L_H(\lambda)}{p(\lambda)}, \quad \mu(\lambda)^2 = Q_4(\lambda).$$

The local phase mismatch between the two colliding transport sheets has the form

$$\mathcal{S}_{+-}(u; H) = \sigma_v(H)(u - u_*)^{3/2} + O((u - u_*)^{5/2}).$$

We use the *linear Langer coordinate*

$$\zeta_v(H; u) = \left(\frac{3i}{4} \sigma_v(H) \right)^{2/3} (u - u_*), \tag{1}$$

so that

$$\frac{i}{2}\mathcal{S}_{+-}(u; H) = \frac{2}{3}\zeta_v(H; u)^{3/2} + O(t^5).$$

The WKB-normalized Airy basis is

$$\psi_{\pm}^{\text{WKB}}(\zeta) = \zeta^{-1/4} \exp\left(\pm \frac{2}{3}\zeta^{3/2}\right).$$

2 The local inverse branches of the common-sheet map

Expand $u(\lambda)$ at v :

$$u(\lambda) = u_* + \frac{u_2}{2!}(\lambda - v)^2 + \frac{u_3}{3!}(\lambda - v)^3 + \frac{u_4}{4!}(\lambda - v)^4 + \frac{u_5}{5!}(\lambda - v)^5 + \cdots,$$

where $u_k = u^{(k)}(v)$. Solving $u - u_* = t^2$ gives two analytic inverse branches

$$\lambda_{\pm}(t) = v \pm c_1 t + c_2 t^2 \pm c_3 t^3 + c_4 t^4 + O(t^5), \quad (2)$$

with

$$c_1 = \sqrt{\frac{2}{u_2}}, \quad c_2 = -\frac{u_3}{3u_2^2}, \quad (3)$$

$$c_3 = \sqrt{\frac{2}{u_2}} \frac{5u_3^2 - 3u_2 u_4}{36u_2^3}, \quad c_4 = \frac{-9u_2^2 u_5 + 45u_2 u_3 u_4 - 40u_3^3}{270u_2^5}. \quad (4)$$

These coefficients depend only on the shared commuting-family geometry.

3 The transport/half-form factor to next nontrivial order

Expand $\Sigma^{(2)}$ at v :

$$\Sigma^{(2)}(\lambda) = \Sigma_1(\lambda - v) + \frac{\Sigma_2}{2!}(\lambda - v)^2 + \frac{\Sigma_3}{3!}(\lambda - v)^3 + \frac{\Sigma_4}{4!}(\lambda - v)^4 + \cdots,$$

where $\Sigma_k = \partial_{\lambda}^k \Sigma^{(2)}(v)$. Substituting (2) gives

$$\Sigma_{\pm}(t) := \Sigma^{(2)}(\lambda_{\pm}(t)) = \pm A_1 t + A_2 t^2 \pm A_3 t^3 + A_4 t^4 + O(t^5), \quad (5)$$

with

$$\begin{aligned} A_1 &= \Sigma_1 c_1, \\ A_2 &= \Sigma_1 c_2 + \frac{\Sigma_2}{2} c_1^2, \\ A_3 &= \Sigma_1 c_3 + \Sigma_2 c_1 c_2 + \frac{\Sigma_3}{6} c_1^3, \\ A_4 &= \Sigma_1 c_4 + \frac{\Sigma_2}{2} (c_2^2 + 2c_1 c_3) + \frac{\Sigma_3}{2} c_1^2 c_2 + \frac{\Sigma_4}{24} c_1^4. \end{aligned} \quad (6)$$

On the t -cover, $\sqrt{du} = \sqrt{2t} \sqrt{dt}$, so the transport/half-form factor is

$$\mathcal{T}_{\pm}(t) := \sqrt{2t} \Gamma_{\pm}(t) = \sqrt{2} \delta_{\pm} t^{1/2} \Sigma_{\pm}(t)^{-1/2}.$$

Set

$$x := \frac{A_2}{A_1}, \quad y := \frac{A_3}{A_1}, \quad z := \frac{A_4}{A_1}.$$

Then the determinant-one normalized transport gauge has the logarithmic expansion

$$\widehat{G}_v^{\text{tr}}(t) = \widehat{G}_v^{\text{lead}} \exp(K_1^{\text{geom}}(v)t\sigma_3 + K_3^{\text{geom}}(v)t^3\sigma_3 + O(t^5)), \quad (7)$$

where

$$K_1^{\text{geom}}(v) = -\frac{A_2}{2A_1} \quad (8)$$

and

$$K_3^{\text{geom}}(v) = -\frac{A_4}{2A_1} + \frac{A_2A_3}{2A_1^2} - \frac{A_2^3}{6A_1^3}. \quad (9)$$

Equivalently,

$$\widehat{G}_v^{\text{tr}}(t) = \widehat{G}_v^{\text{lead}} \left[I + K_1^{\text{geom}}(v)t\sigma_3 + \frac{K_1^{\text{geom}}(v)^2}{2}t^2I + K_3^{\text{geom}}(v)t^3\sigma_3 + O(t^4) \right].$$

Proposition 1 (Closed symbolic transport coefficients). *In the linear Langer/WKB normalization, the first two nontrivial transport/half-form gauge coefficients are geometry-only and are given by (8) and (9). In particular,*

$$K_1^{\text{geom}}(v) = \sqrt{\frac{2}{u''(v)}} \left(\frac{u'''(v)}{12u''(v)} - \frac{\Sigma^{(2)''}(v)}{4\Sigma^{(2)'}(v)} \right). \quad (10)$$

Proof. The formulas for $A_1, \dots, A_4$ come from substituting (2) into the Taylor series of $\Sigma^{(2)}$. The expansion of $\mathcal{T}_{\pm}$ is obtained from the binomial series for $(1+z)^{-1/2}$. Writing the determinant-one normalized diagonal ratio in exponential form gives

$$\frac{1}{2} \log \frac{\mathcal{T}_+(t)}{\mathcal{T}_-(t)} = -\frac{x}{2}t + \left(-\frac{x^3}{6} + \frac{xy}{2} - \frac{z}{2} \right) t^3 + O(t^5),$$

which yields (8) and (9). Substituting the expressions for A_1 and A_2 into (8) gives (10) directly. $\square$

4 The phase part and the first appearance of member dependence

Expand the phase carrier difference along the two branches:

$$\Delta_H(\lambda_+(t)) - \Delta_H(\lambda_-(t)) = 2D_1(H)t + 2D_3(H)t^3 + O(t^5), \quad (11)$$

where

$$D_1(H) = c_1\Delta'_H(v), \quad D_3(H) = c_3\Delta'_H(v) + c_1c_2\Delta''_H(v) + \frac{c_1^3}{6}\Delta'''_H(v). \quad (12)$$

Since $du = 2t dt$, integrating gives

$$\mathcal{S}_{+-}(t; H) = \frac{4}{3}D_1(H)t^3 + \frac{4}{5}D_3(H)t^5 + O(t^7). \quad (13)$$

By definition of the local cubic coefficient,

$$\sigma_v(H) = \frac{4}{3}D_1(H).$$

Using the linear Langer coordinate (1), one has

$$\frac{2}{3}\zeta_v(H;t)^{3/2} = \frac{i}{2}\sigma_v(H)t^3.$$

Therefore

$$\frac{i}{2}\mathcal{S}_{+-}(t;H) - \frac{2}{3}\zeta_v(H;t)^{3/2} = \Xi_5(H)t^5 + O(t^7), \quad \boxed{\Xi_5(H) = \frac{2i}{5}D_3(H)}. \quad (14)$$

Proposition 2 (First member-dependent term). *In the linear Langer/WKB normalization, the member-dependent phase remainder begins at order t^5 and is given by (14). Equivalently,*

$$\Phi_v^{\text{ph}}(t;H) = \Xi_5(H)t^5 + O(t^7).$$

No member-dependent contribution appears at orders t or t^3 .

Proof. Because the two colliding branches are exchanged by $t \mapsto -t$, the phase-carrier difference is an odd function of t , hence has the form (11). Integrating against $du = 2t dt$ produces only odd powers $t^3, t^5, \dots$ in (13). The linear Langer coordinate is chosen precisely so that the cubic term is removed. The first surviving coefficient is therefore the t^5 term, namely $\Xi_5(H)$. $\square$

5 Closed-form transport/phase factorization

Let the exact local half-form amplitudes be

$$\Psi_{\pm}(t;H) = \sqrt{2t} \Gamma_{\pm}(t) R_{\pm}(t;H) \sqrt{dt}.$$

Write the reduced sheetwise carriers as

$$R_{\pm}(t;H) = \exp(\Phi_{\text{av}}(t;H)) \exp(\pm \Phi_{\text{rel}}(t;H)), \quad \Phi_{\text{rel}}(t;H) = \frac{i}{2}\mathcal{S}_{+-}(t;H).$$

Then, after removing the common scalar phase Φ_{av} , the determinant-one local gauge admits the factorization

$$\boxed{\widehat{G}_v(t;H) = \widehat{G}_v^{\text{tr}}(t) \exp(\Phi_v^{\text{ph}}(t;H) \sigma_3)}, \quad (15)$$

with

$$\Phi_v^{\text{ph}}(t;H) = \Xi_5(H)t^5 + O(t^7), \quad (16)$$

and $\widehat{G}_v^{\text{tr}}(t)$ given by (7). Therefore

$$\boxed{\widehat{G}_v(t;H) = \widehat{G}_v^{\text{lead}} \exp(K_1^{\text{geom}}(v)t \sigma_3 + K_3^{\text{geom}}(v)t^3 \sigma_3 + O(t^5)) \exp(\Xi_5(H)t^5 \sigma_3 + O(t^7))}. \quad (17)$$

Theorem 1 (Transport versus phase in the local gauge expansion). *In the linear Langer/WKB normalization at a simple transport ramification point v , the determinant-one local gauge separates into a purely transport/half-form part and a purely phase/member-dependent part as in (17). The coefficients $K_1^{\text{geom}}(v)$ and $K_3^{\text{geom}}(v)$ depend only on the shared commuting-family geometry at v , while the first genuine member-dependent coefficient is $\Xi_5(H)$ and appears at order t^5 .*

Proof. Combine Proposition 1 with Proposition 2. The transport/half-form part is built entirely from the shared quantities $u(\lambda)$ and $\Sigma^{(2)}(\lambda)$, while the phase remainder is built from the derivatives of $\Delta_H(\lambda)$ and therefore carries the member dependence. The claimed orders follow from the explicit expansions. $\square$

Remark 1 (A nonlinear normal-form caveat). *The statement that member dependence first appears at order t^5 is tied to the linear Langer normalization (1). If one allows a nonlinear redefinition*

$$\zeta = \kappa_H t^2 (1 + \ell_H t^2 + \cdots),$$

then part of the t^5 phase remainder can be absorbed into the coordinate choice. What is invariant is the structural split between the geometry-only transport piece and the phase/member-dependent piece.

6 Consequences for the local jump constants

Fix the upper-half-plane simple transport ramification point and the local selector crossing

$$S_1 \longrightarrow S_0.$$

In the normalized Airy model, the corresponding local jump matrix is

$$J_{1 \rightarrow 0}^{\text{norm}} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} = I - E_{21}, \quad E_{21} := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \quad (18)$$

At the level of the proved gauge structure, the determinant-one exact local gauge is diagonal through the displayed orders:

$$\hat{G}_v(t; H) = \hat{G}_v^{\text{lead}} \exp(K_v^{\text{geom}}(t) \sigma_3) \exp(\Phi_v^{\text{ph}}(t; H) \sigma_3),$$

where

$$K_v^{\text{geom}}(t) = K_1^{\text{geom}}(v)t + K_3^{\text{geom}}(v)t^3 + O(t^5), \quad \Phi_v^{\text{ph}}(t; H) = \Xi_5(H)t^5 + O(t^7).$$

Because diagonal conjugation preserves the lower-unitriangular rank-one form, the exact local selector block stays in the same one-parameter family as the Airy block.

Corollary 1 (Scalar jump law through order t^5). *For the upper-half-plane $S_1 \rightarrow S_0$ crossing, the exact local selector jump has the form*

$$\boxed{J_v^{\text{exact}}(t; H) = I + \mathfrak{s}_v(t; H) E_{21}}, \quad (19)$$

with scalar jump coefficient

$$\boxed{\mathfrak{s}_v(t; H) = -i \exp\left(2K_1^{\text{geom}}(v)t + 2K_3^{\text{geom}}(v)t^3 + 2\Xi_5(H)t^5 + O(t^7)\right)}. \quad (20)$$

Equivalently,

$$\boxed{\log\left(\frac{\mathfrak{s}_v(t; H)}{-i}\right) = 2K_1^{\text{geom}}(v)t + 2K_3^{\text{geom}}(v)t^3 + 2\Xi_5(H)t^5 + O(t^7)}. \quad (21)$$

At the conjugate lower-half-plane point,

$$\boxed{J_{\bar{v}}^{\text{exact}}(t; H) = I + \mathfrak{s}_{\bar{v}}(t; H) E_{21}, \quad \mathfrak{s}_{\bar{v}}(t; H) = +i \exp\left(2\overline{K_1^{\text{geom}}(v)}t + 2\overline{K_3^{\text{geom}}(v)}t^3 + 2\overline{\Xi_5(H)}t^5 + O(t^7)\right).} \quad (22)$$

In particular, the local jump constant is geometry-only through order t^3 , and the first genuine member dependence appears at order t^5 .

Proof. Let

$$D_v(t; H) := \hat{G}_v^{\text{lead}} \exp(K_v^{\text{geom}}(t)\sigma_3) \exp(\Phi_v^{\text{ph}}(t; H)\sigma_3) = \begin{pmatrix} d_+(t; H) & 0 \\ 0 & d_-(t; H) \end{pmatrix}.$$

Then

$$J_v^{\text{exact}}(t; H) = D_v(t; H)^{-1} J_{1 \rightarrow 0}^{\text{norm}} D_v(t; H) = I - \frac{d_+(t; H)}{d_-(t; H)} E_{21}.$$

Since

$$\frac{d_+(t; H)}{d_-(t; H)} = \exp\left(2K_v^{\text{geom}}(t) + 2\Phi_v^{\text{ph}}(t; H)\right) \times \frac{e^{i\pi/4}}{e^{-i\pi/4}} = i \exp\left(2K_1^{\text{geom}}(v)t + 2K_3^{\text{geom}}(v)t^3 + 2\Xi_5(H)t^5 + O(t^7)\right),$$

the coefficient of E_{21} is the claimed scalar $\mathfrak{s}_v(t; H)$. The logarithmic form is immediate. The lower-half-plane expression follows by conjugation. $\square$

Remark 2 (Why the jump computation simplifies). *The local selector problem has collapsed from a generic 2×2 matching problem to a one-scalar renormalization of the universal Airy jump. The exact local data needed by the global continuation word are therefore:*

1. the active selector point v and the sector crossing $S_s \rightarrow S_{s'}$;
2. the geometry-only coefficients $K_1^{\text{geom}}(v)$ and $K_3^{\text{geom}}(v)$;
3. if higher precision is needed, the first phase/member correction $\Xi_5(H)$.

Everything else is already fixed by the universal Airy model and the conjugate-pair symmetry of the upper/lower ramification points.

Remark 3 (Scope of the corollary). *The formulas above are exact to the displayed orders of the proved gauge expansion. They use only the established diagonal transport/phase factorization. What is still not proved is that no additional off-diagonal or sector-dependent local gauge terms appear before higher order. So the one-scalar law (20) should be read as the strongest current structural statement for the exact local jump, not yet as a fully closed all-orders theorem.*

7 Canonical example

Take the same canonical commuting family used in the previous notes:

$$(\epsilon_0, \epsilon_1, \epsilon_2) = (-2, 0, 3), \quad (\gamma_0, \gamma_1, \gamma_2) = \left(1, \frac{4}{5}, \frac{6}{5}\right).$$

Choose the upper-half-plane ramification point

$$v = -0.8019106146 - 1.1076910061i, \quad u_* = u(v) = -0.1735624567 + 0.8968618574i.$$

Then

$$\begin{aligned} u''(v) &= 0.1611148247 + 0.5485768616i, & u'''(v) &= 1.7051645913 - 0.7517504141i, \\ \Sigma^{(2)'}(v) &= -0.1611148247 - 0.5485768616i, & \Sigma^{(2)''}(v) &= -1.7051645913 + 0.7517504141i. \end{aligned}$$

Substituting into (10) gives

$$K_1^{\text{geom}}(v) = 0.7088279025 + 0.7278828796i.$$

Using the next jet coefficients one finds

$$K_3^{\text{geom}}(v) = -0.0482124825 + 0.9683460218i.$$

For the two members from the selector note,

$$H_{\text{wall}} : \alpha = 0.1, \beta = 0.0432447842161505, \quad H_{\text{off}} : \alpha = 0.1, \beta = 0.08,$$

the phase coefficients are

$$\Xi_5(H_{\text{wall}}) = 0.0511662459 + 0.1064888853i,$$

$$\Xi_5(H_{\text{off}}) = 0.0535527145 + 0.1146585773i.$$

So the transport coefficients $K_1^{\text{geom}}(v)$ and $K_3^{\text{geom}}(v)$ are identical for both members, while the first phase coefficient $\Xi_5(H)$ already distinguishes them.

8 Interpretation

The symbolic picture is now quite sharp.

- The transport/half-form side is organized by the shared local jet of the family-invariant map $u = n/p$ and the family-invariant transport normalization $\Sigma^{(2)} = W_4/p^2$.
- The chosen member H enters through the phase carrier $\Delta_H = \mu L_H/p$.
- The leading cubic phase term is normalized away by the Langer coordinate.
- The first two nontrivial transport coefficients are therefore geometry-only.
- The first unavoidable member dependence in the linear normalization appears at order t^5 .

This is the clean symbolic explanation for the numerical observation that nearby members of the same commuting family can have very similar local bounded-remainder behavior even when their selector activation differs.